



OBJECTIVES of Our Solution:

KEY FEATURES

Automatically analyzes every HYPE app review with EDA to discover patterns

Detects **negative** topics (bugs, complaints, usability issues)

Identifies **recurring problems** and emerging trends

Tags each review **consistently** and at scale

BENEFITS

Faster insights for the Research & Customer Insight team

Reduced manual reading and tagging **effort**

Summarizes large volumes of feedback in seconds

Improved visibility on customer **pain points**

AI TOOL FOR FEEDBACK AUTOMATION



Defining the VALUE PROPOSITION

8 DECENT WORK AND ECONOMIC GROWTH



By using automated NLP, we enhance feedback analysis **efficiency**, reduce **manual effort**, and **accelerate** decision-making. Real-time insights from user reviews enable data-driven product improvements, **driving growth** and fostering a **more sustainable** business environment.

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE



By utilizing cutting-edge **AI technology**, we automate topic detection, improving **operational efficiency** and modernizing product development processes. This supports the growth of a more **data-driven** and **technologically advanced infrastructure**.



HYPOTHESIS AND RESEARCH QUESTIONS

Driving Our Investigation



- **RQ1** — Does machine translation affect the semantic quality of topic detection on multilingual app reviews?
- **RQ2** — How do different stopwords-removal strategies affect the performance of embedding-based topic modeling on app reviews?
- **RQ3** — Which configuration of BERTopic's modular components (embeddings, dimensionality reduction, clustering, and topic representation) yields the best topic quality?
- **RQ4** — Do semi-supervised topic models outperform unsupervised approaches in identifying relevant issues from app reviews?
- **RQ5** — Do embedding-based semantic similarity methods provide reliable mappings between data-driven topics and the existing taxonomy of app issues?

From Tasks to **DEVELOPMENT CHOICES**



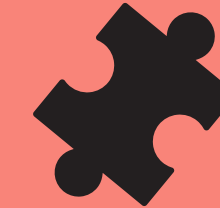
1. EDA & Pre-processing

Exploratory data analysis and preprocessing of text and features.



2. Topic Detection

Detecting topics and comparing them with the HYPE taxonomy.



3. Extra Tasks

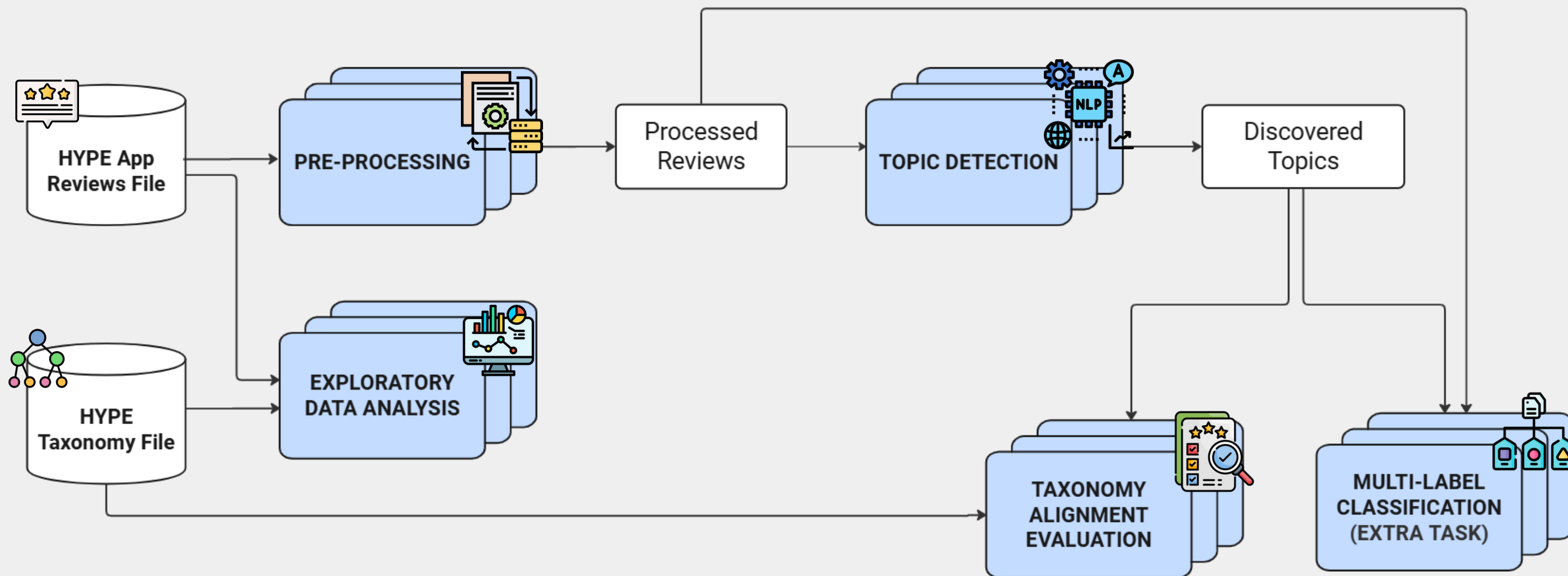
Future work: multi-label classification and hierarchical topic detection.

We translated the three main project tasks into a concrete, reproducible development pipeline by defining clear **development choices** for each stage of the workflow...



FUNCTIONAL DIAGRAM

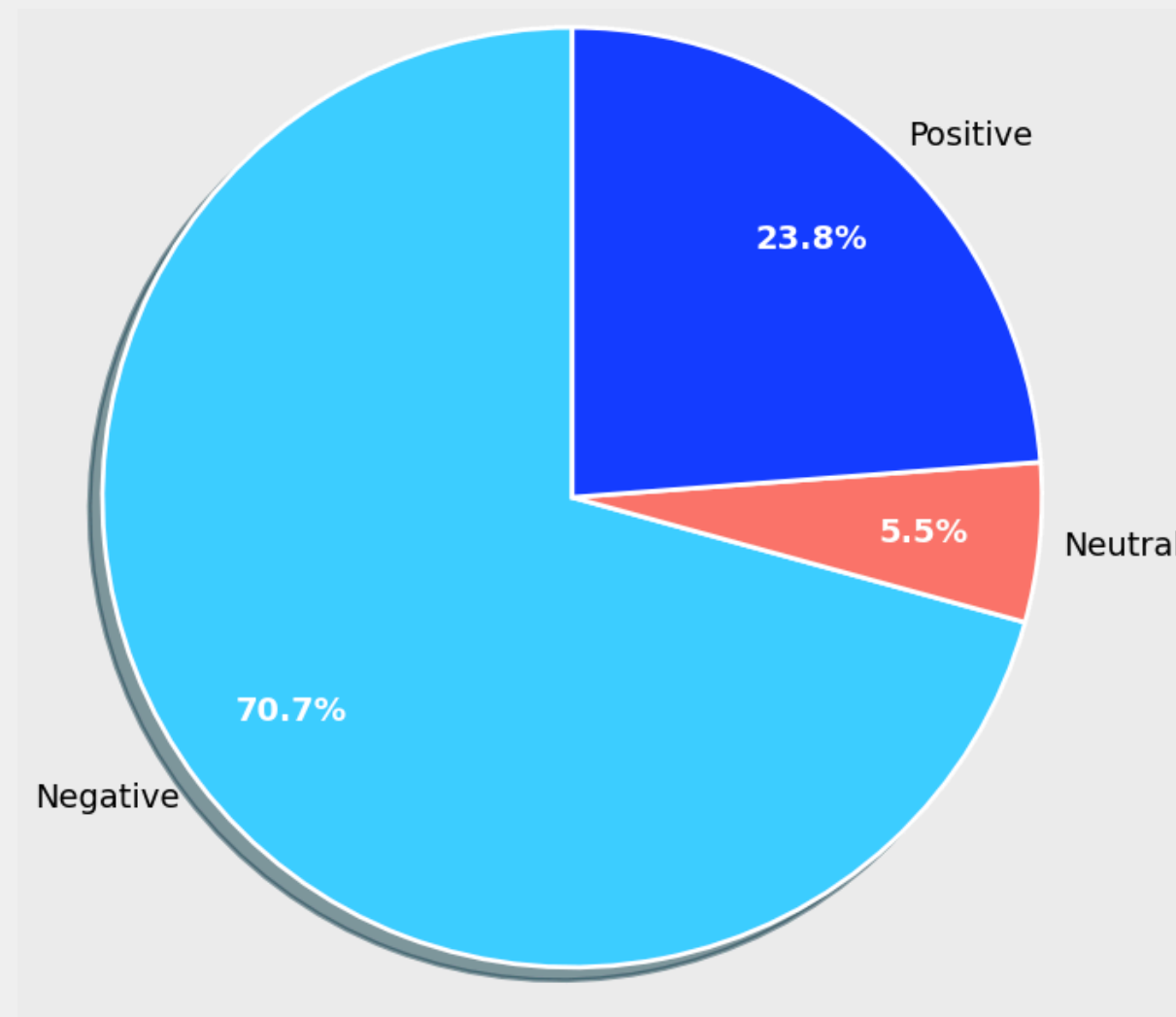
UPDATE



Introducing the **DATASET**

id_review	review	sentiment	rating	labels	is_labels_	model_version	review_date
00000038-3b3b-5102-be9a-7cf800000000	Nell'applicazione (ricarica conto) non ho più la possibilità di ricaricare in contanti non esiste più l'opzione è cambiato qualcosa?	Negative	5	["problemi_ricarica"]	FALSO	1.0.0	01/10/2024

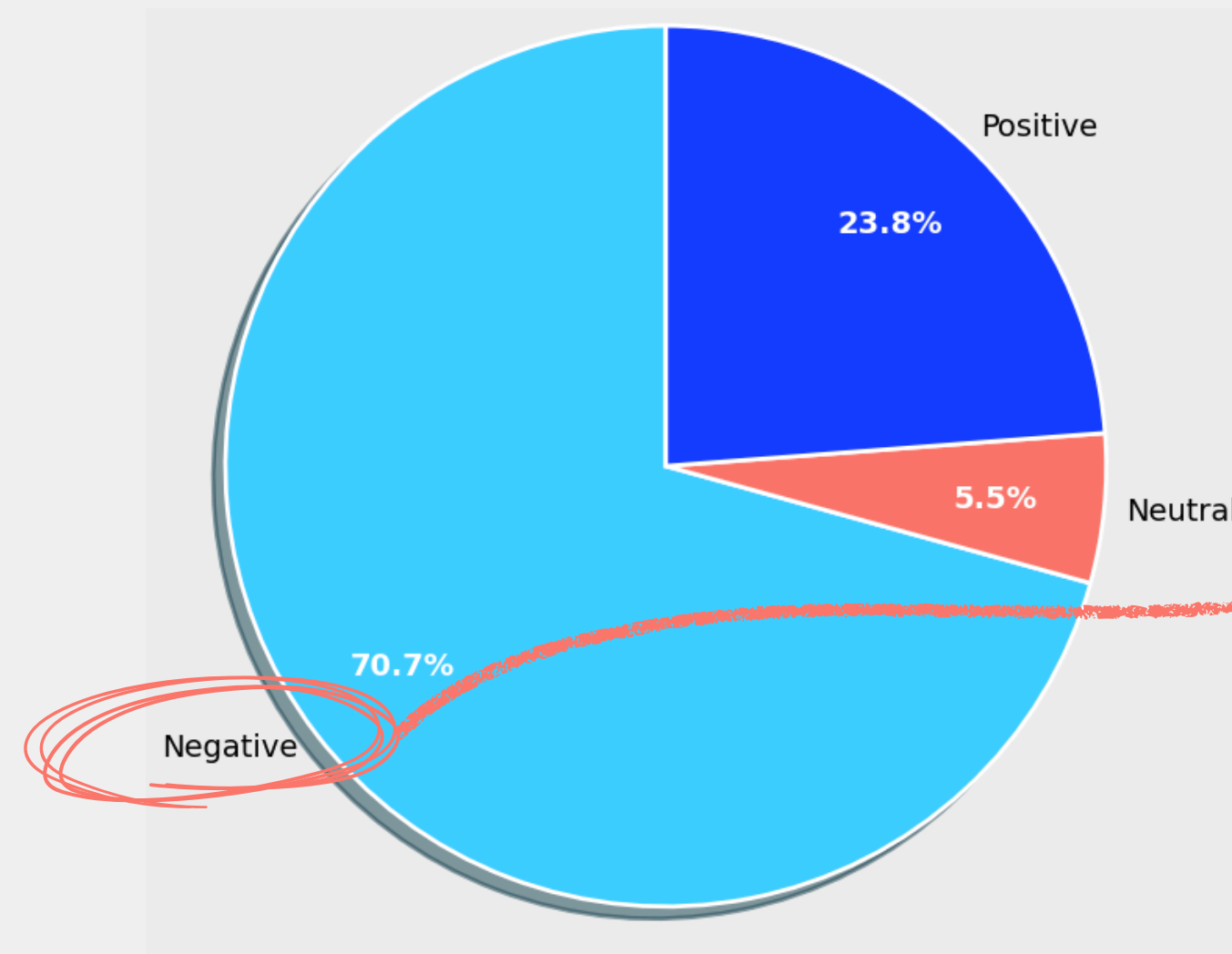
Our data resource consists of
6,150 records, covering,
nowdays, **20 different labels**
detected for analysis.



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Our data resource consists of **6,150 records**, covering, nowadays, **20 different labels** detected for analysis.



We need to focus specifically on the **negative reviews**, as they allow us to identify critical feedback and the main areas that require improvement

Introducing the DATASET

The **sentiment label** is used to select the appropriate subset of reviews, allowing us to prioritize the negative feedback.

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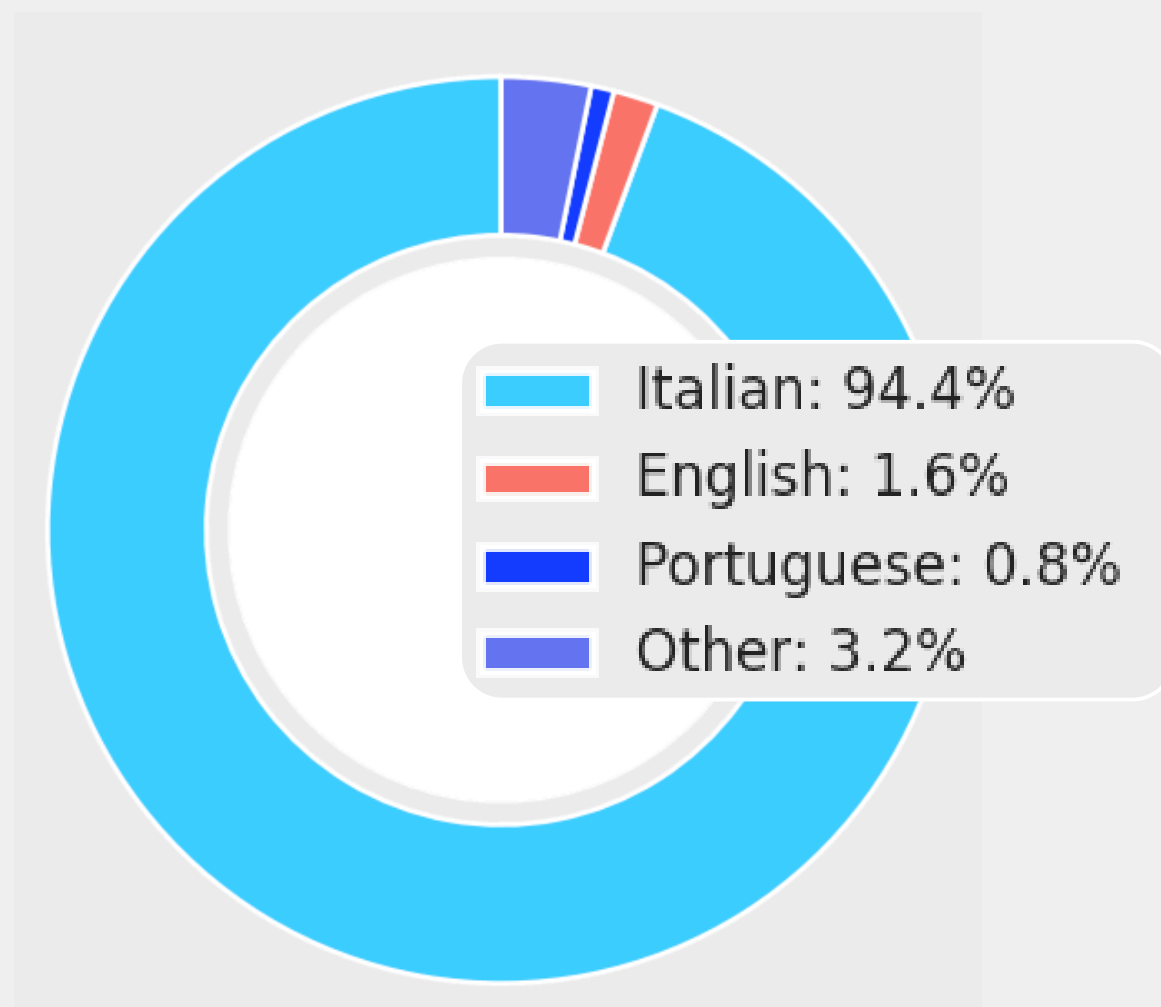
The **review text** plays a central role in our analysis, as all processing steps are applied directly to it in order to identify the underlying topic.

The **label** enables a direct comparison between our model's output and the one generated by the company's previously adopted system.

From the provided initial dataset, we performed **a series of transformations** to refine it...



How Do We Handle **MULTILINGUAL REVIEWS** in the dataset?



Some reviews are written in languages **other than Italian**. To ensure consistent analysis, all non-Italian reviews are automatically **translated into Italian** using NLLB.

A **NLLB** – No Language Left Behind

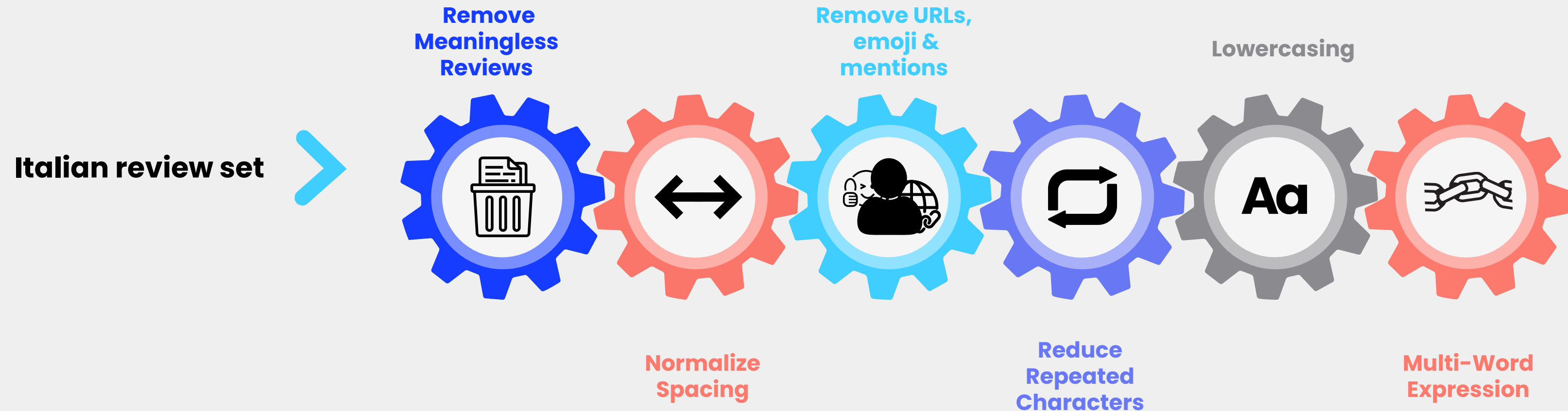
worst app 👎



l'app peggiore 👎

Each review is first processed through a dedicated **transformer-based language-detection model** to identify its source language. Then, the text is translated using **NLLB**, a state-of-the-art neural machine translation model that provides highly reliable translations across a wide range of languages.

PRE-PROCESSING Workflow



How Do We Interpret **NEUTRAL REVIEWS**?

Apro l'app più volte a settimana e ogni volta mi chiede di inserire la password e poi parte il Face ID.

Vi prego fate qualcosa

Neutral

sì deve sempre migliorare

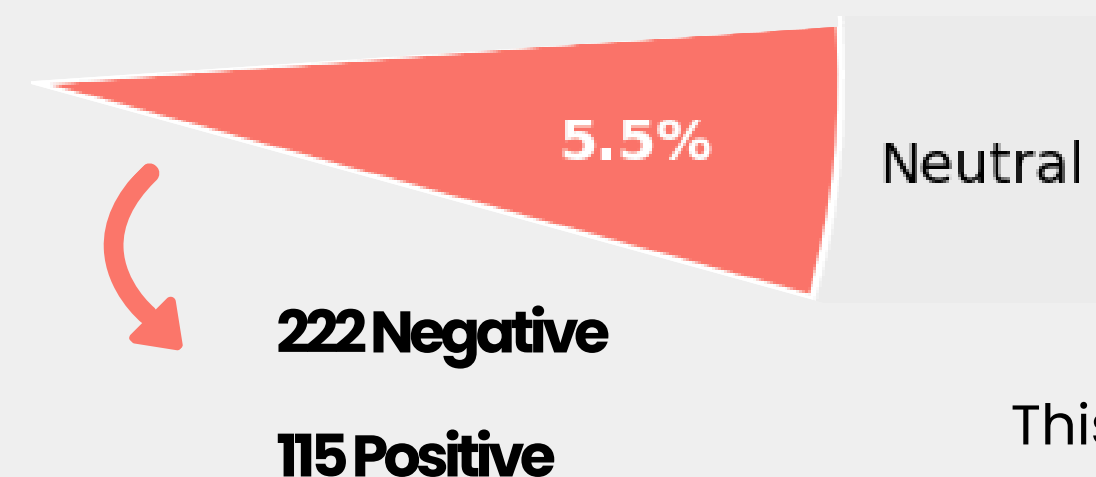
Neutral

Neutral ≠ always problem-free

The initial neutral label does not always reflect the true nature of user feedback. This inconsistency motivates our dataset refinement. We therefore performed a **sentiment-analysis step** using two models and aggregated their outputs to obtain a more reliable sentiment prediction.

🤗 XLM-R

🤗 FEEL-IT



This analysis produced a total of **4571 negative** reviews to focus on.

How Do We Deal With **REDUNDANT REVIEWS** ?

Review 1: Chiuderò il conto Hype. Non proteggono dalle frodi, dicendo di bloccare la carta e di emetterne una nuova e a pagamento
Review 2: Non proteggono dalle frodi, dicendo di bloccare la carta e di emetterne una nuova e a pagamento

TF-IDF

A lexical representation based on word frequency; it compares texts using exact terms but cannot capture meaning or paraphrases.

SBERT

A semantic embedding model that captures sentence meaning, enabling similarity detection even when different words are used.



E5

A modern multilingual embedding model optimized for semantic similarity and retrieval, offering more robust and consistent representations than SBERT.

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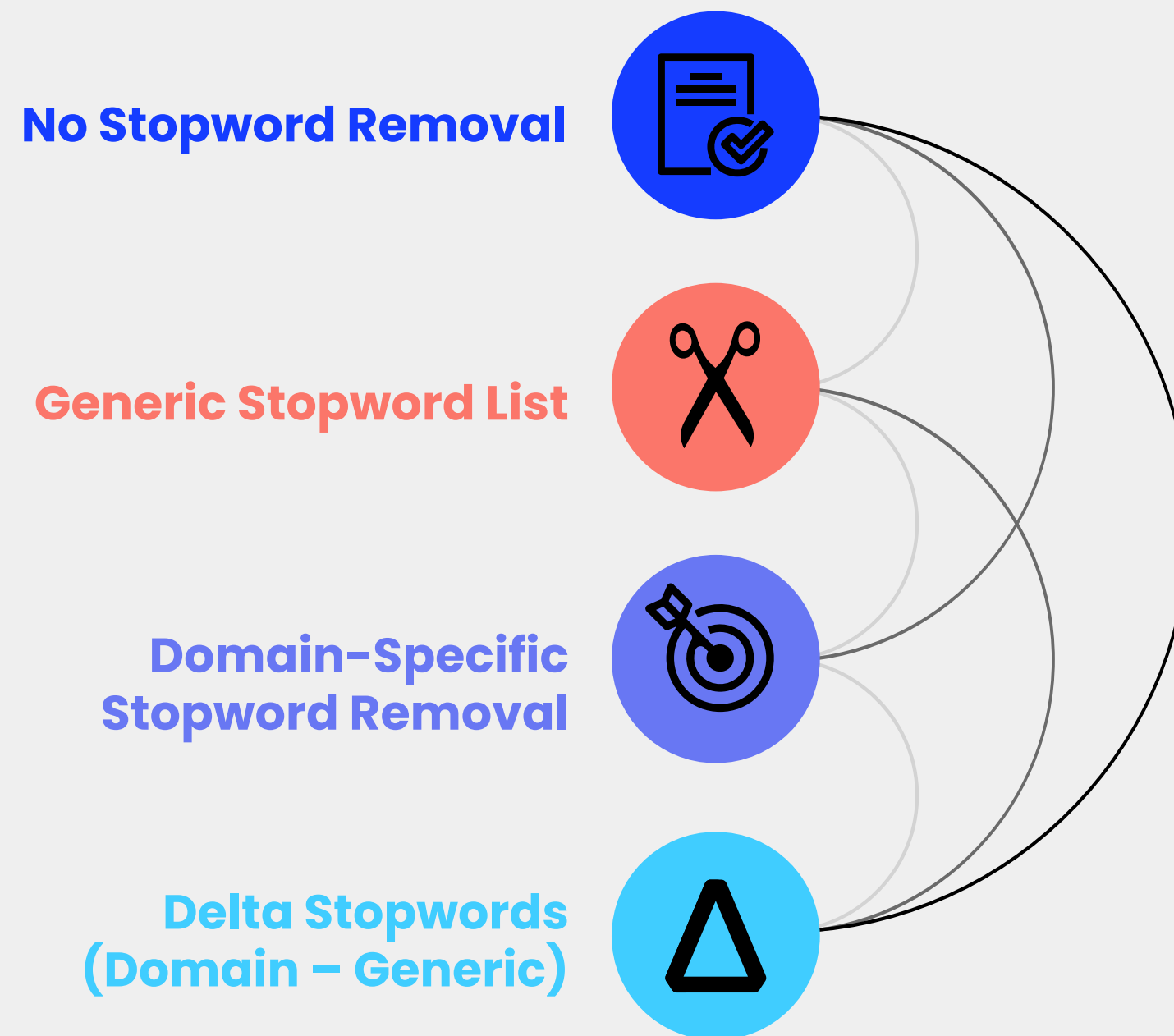
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TF-IDF

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After comparing TF-IDF, SBERT, and E5, we opted for a basic **TF-IDF approach**: while the embedding models performed well semantically, they reduced the dataset too aggressively, making TF-IDF the most suitable choice for preserving coverage. This analysis reduce our dataset to **4454 negative** reviews.

Comparative Evaluation of **STOPWORD STRATEGIES**



Literature shows that traditional stopwords removal is often **sub-optimal** for embedding-based models, since these models rely on the full syntactic structure of the input text. In our study, we want to evaluate the impact of different stopwords strategies, including a domain-specific stopwords list generated in a data-driven way using another banking app reviews dataset, to assess whether domain knowledge can **provide benefits**.

Understanding TOPIC DETECTION



DEFINITION

An **unsupervised NLP** method that models the latent semantic structure of a text corpus by grouping semantically similar texts



OBJECTIVE

Identify a **set of distinct topics** defined by high-relevance **keywords** and the **texts** most aligned with each topic



APPROACH

Convert text into **numerical representations**, organize them into **latent semantic groups** through clustering or probabilistic models, and extract the **salient terms** defining each topic.



TECHNIQUES

LDA, **NMF**, **LSA**, **Top2Vec**, and **BERTopic**

Comparing MODELS

LDA

Probabilistic document as a mixture of topics, topic as a distribution over words

PROS: ✓ SIMPLE, FAST, INTERPRETABLE

CONS: ✗ NO SEMANTICS, FIXED #TOPICS, WEAK ON SHORT TEXT

NMF

Matrix factorization of the document-term matrix into topic components.

PROS: ✓ COHERENT TOPICS, STABLE, GOOD ON SHORT TEXT

CONS: ✗ NO SEMANTICS, FIXED #TOPICS

BERTopic

Embedding-based model that uses transformer representations, HDBSCAN clustering and c-TF-IDF keyword extraction.

PROS

✓ HIGH **SEMANTIC UNDERSTANDING**

✓ POWERFUL ALSO ON **SHORT AND NOISY TEXT**

✓ DOES NOT REQUIRE STRONG PRE-PROCESSING

CONS

✗ COMPUTATIONALLY HEAVY

✗ SENSITIVE TO EMBEDDING MODEL CHOICE

✗ PRODUCES A SINGLE-TOPIC ASSIGNMENT

Comparing MODELS

LDA

Probabilistic document as a mixture of topics, each topic is a distribution over words

PROS: SIMPLE, FAST
CONS: ✗ POOR SEMANTIC UNDERSTANDING, FIXED #TOPICS, WEAK ON SHORT TEXT

BASELINE

TF-IDF

Matrix factorization of the document-term matrix into topic components

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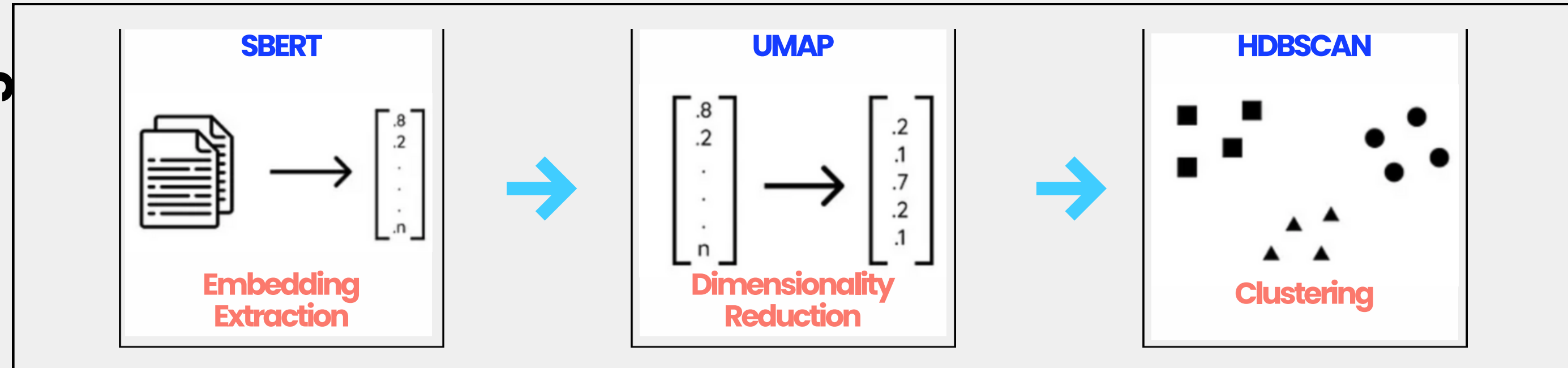
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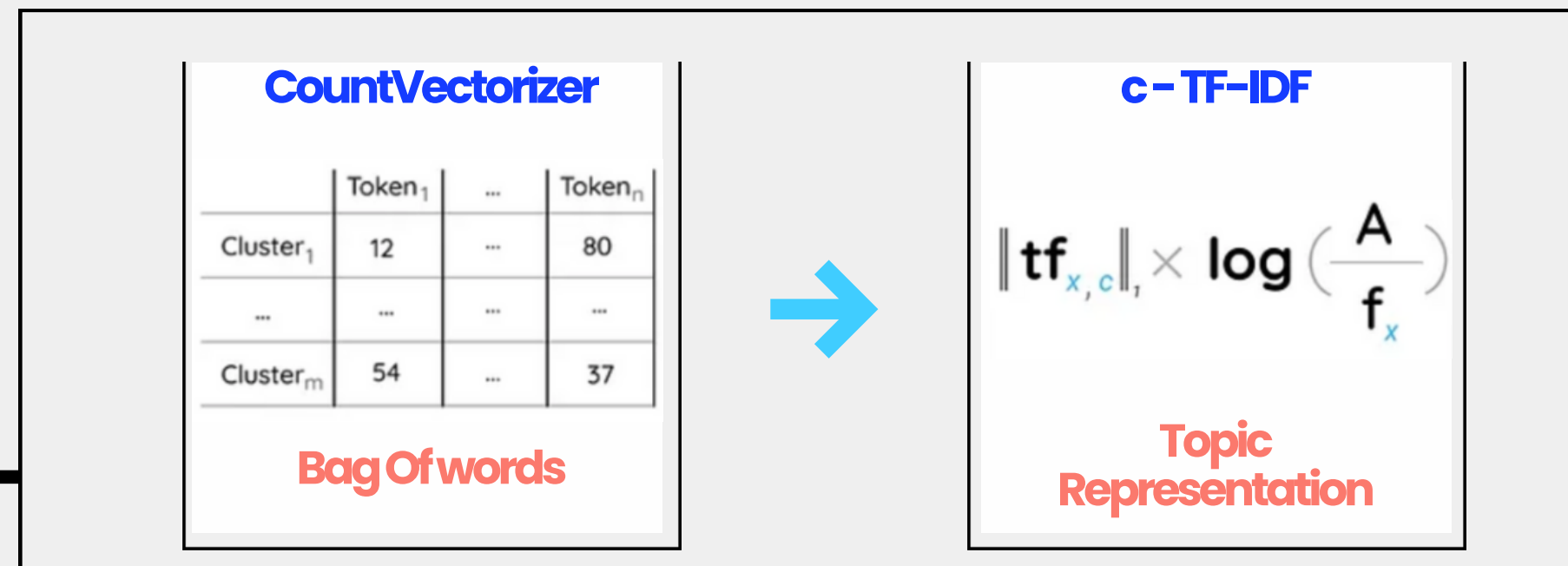
BERTopic

BERTopic

Clustering

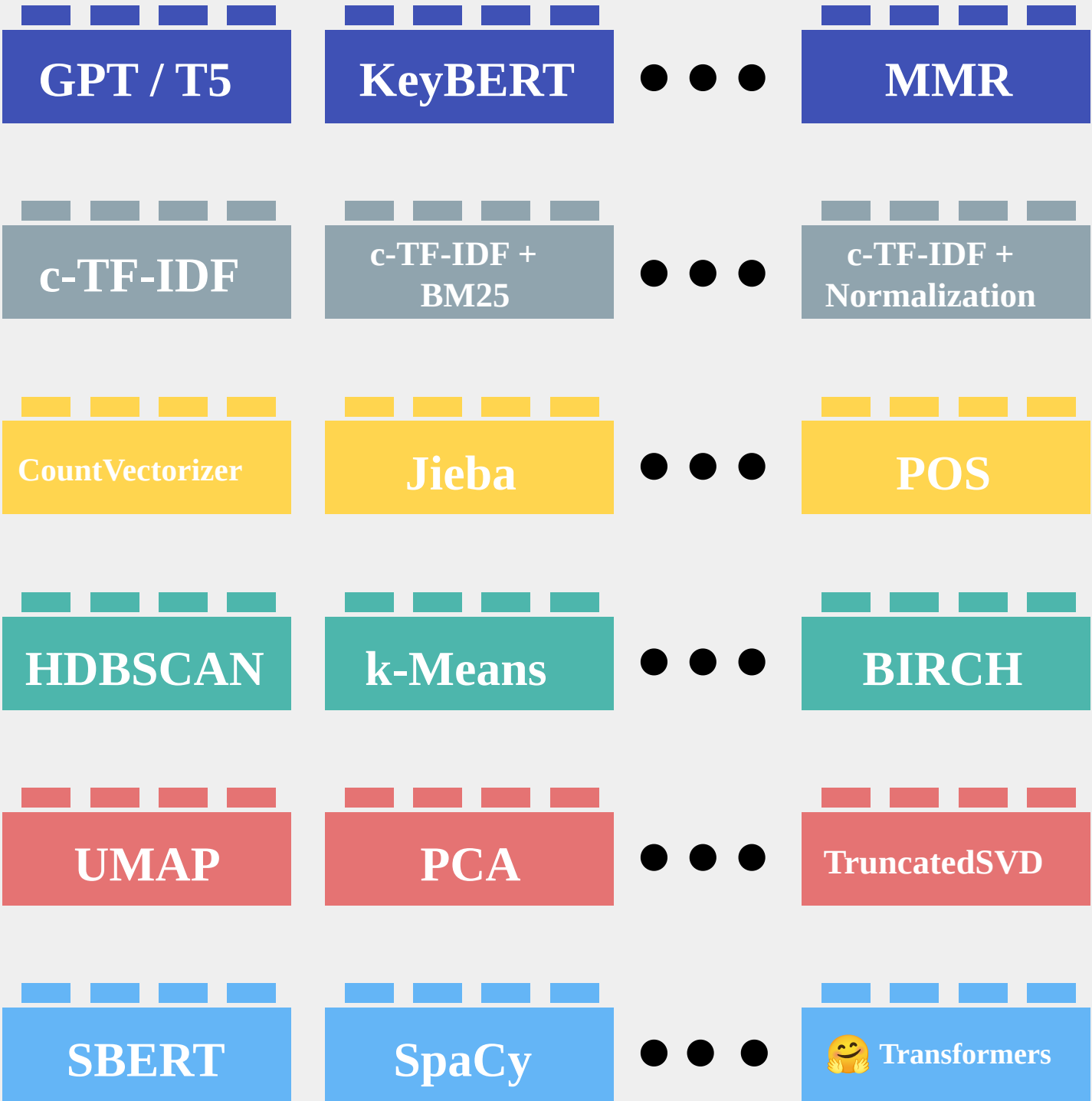


Topic Extraction

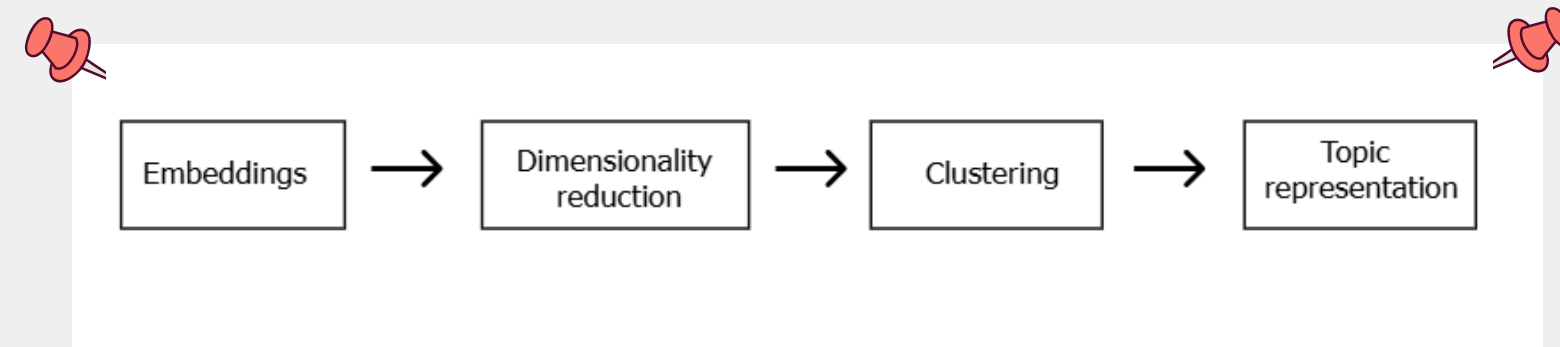


BERTopic

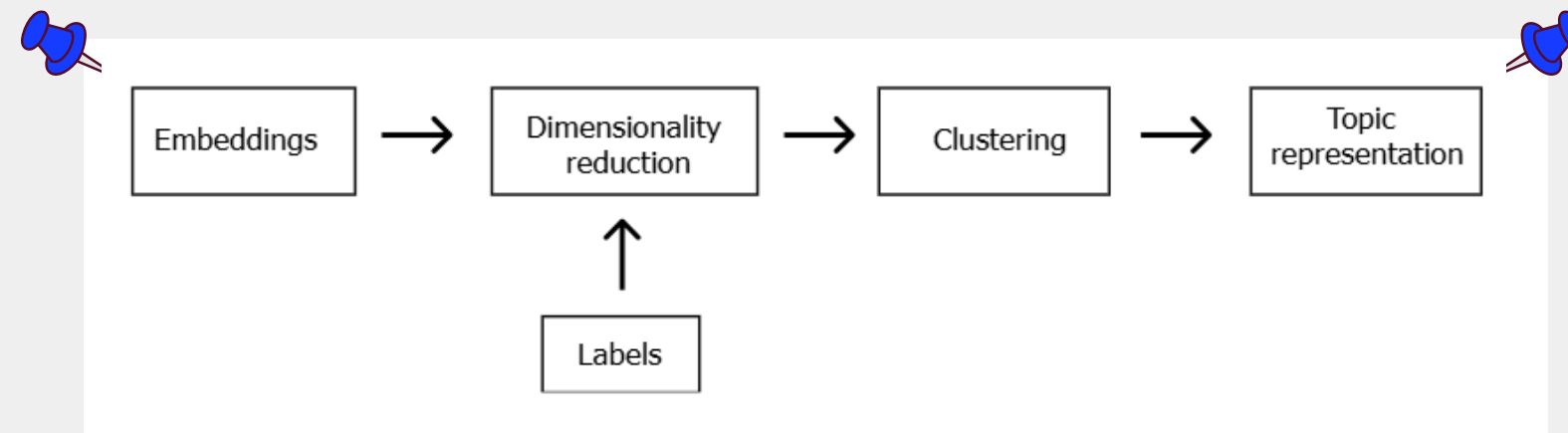
Offers a completely **MODULAR ARCHITECTURE**



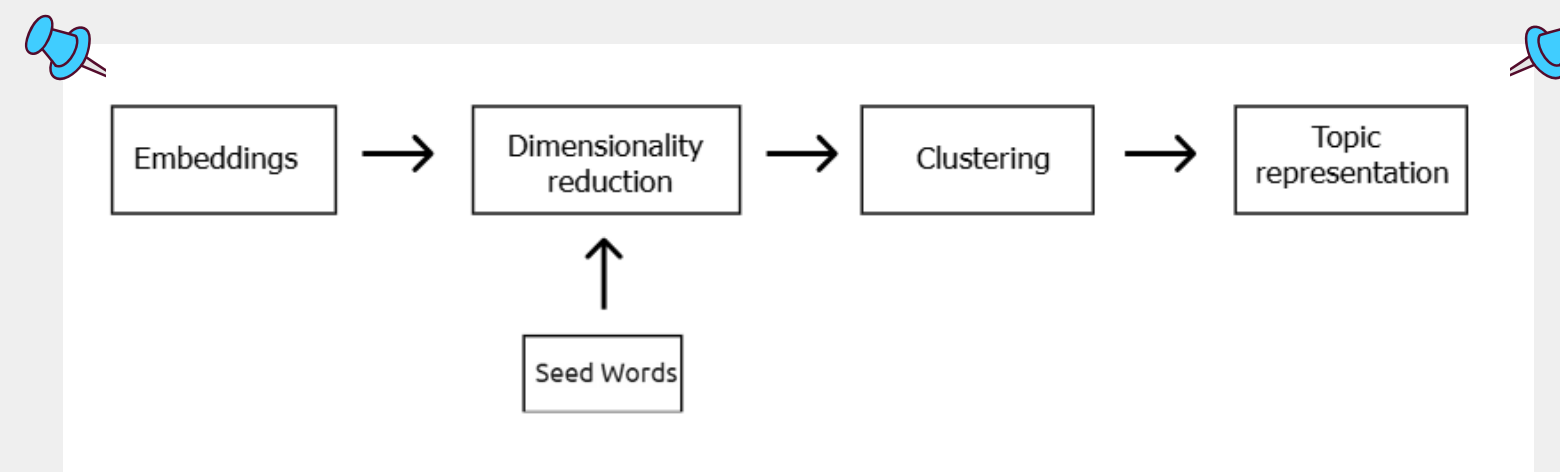
Topic Modeling: THREE APPROACHES COMPARED



**TRADITIONAL TOPIC MODELING
(UNSUPERVISED)**



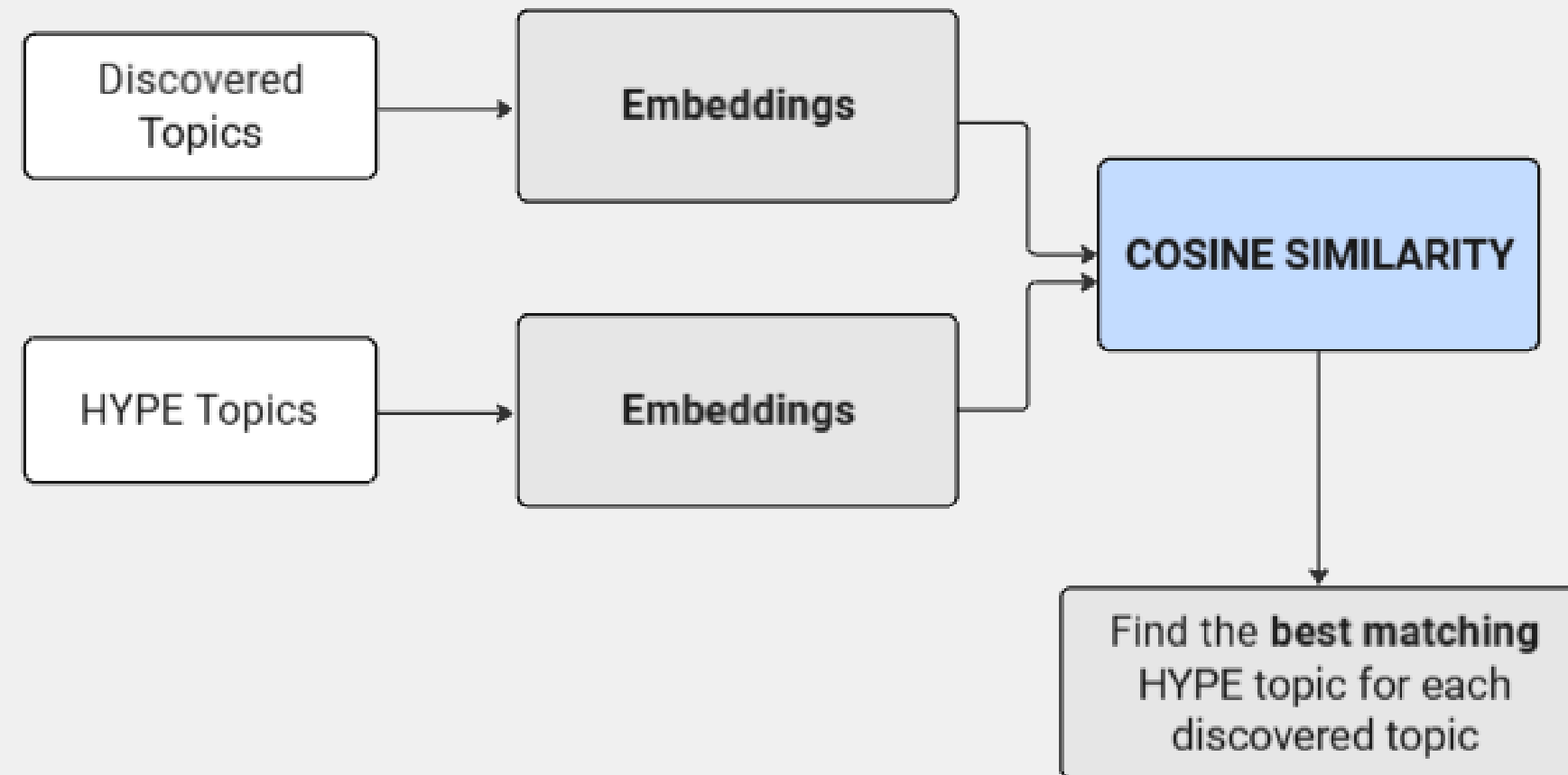
SEMI-SUPERVISED TOPIC MODELING



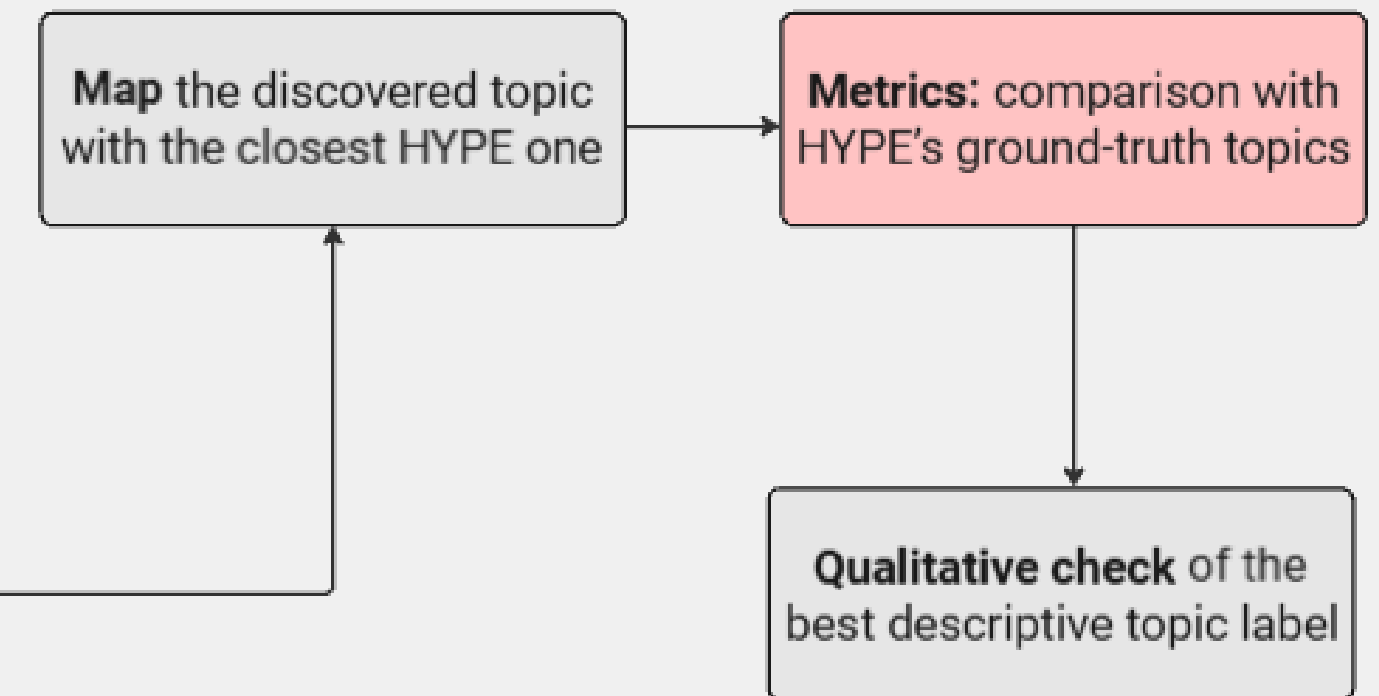
GUIDED TOPIC MODELING

Aligning Topics to the HYPETAXONOMY

TAXONOMY ALIGNMENT

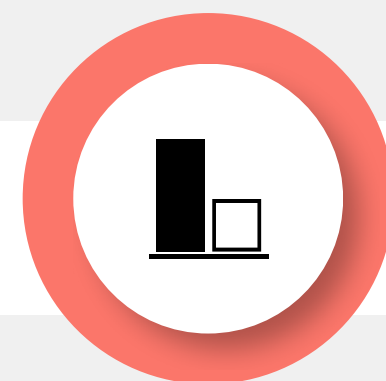


EVALUATION AND COMPARISON



TOPIC DETECTION Workflow

Pre-processed
reviews



Apply LDA as a
baseline



Apply BERTopic
trying different
architecture
combinations



Test Different
Topic Detection
approaches



Taxonomy
Alignment

EXTRA TASKS

Multi-Label Classification (Optional Extension)



Some reviews mention more than one issue simultaneously, making single-label topic assignment insufficient.

➤ To be implemented if **time** allows!

Da oggi sto riscontrando problemi ad accedere al mio account, sia da App che da Web. Sono praticamente tagliata fuori dal mio conto e non riesco a ricevere assistenza. Nella chat con operatore sto scrivendo e nessuno risponde. Questa cosa è pessima

Login Issues

Customer Support Issues

How we work as a **TEAM**



M
A
N
A
G
E





Refinement of the **functional diagram**, application of **machine translation**, implementation of standard **preprocessing** and general research activities.

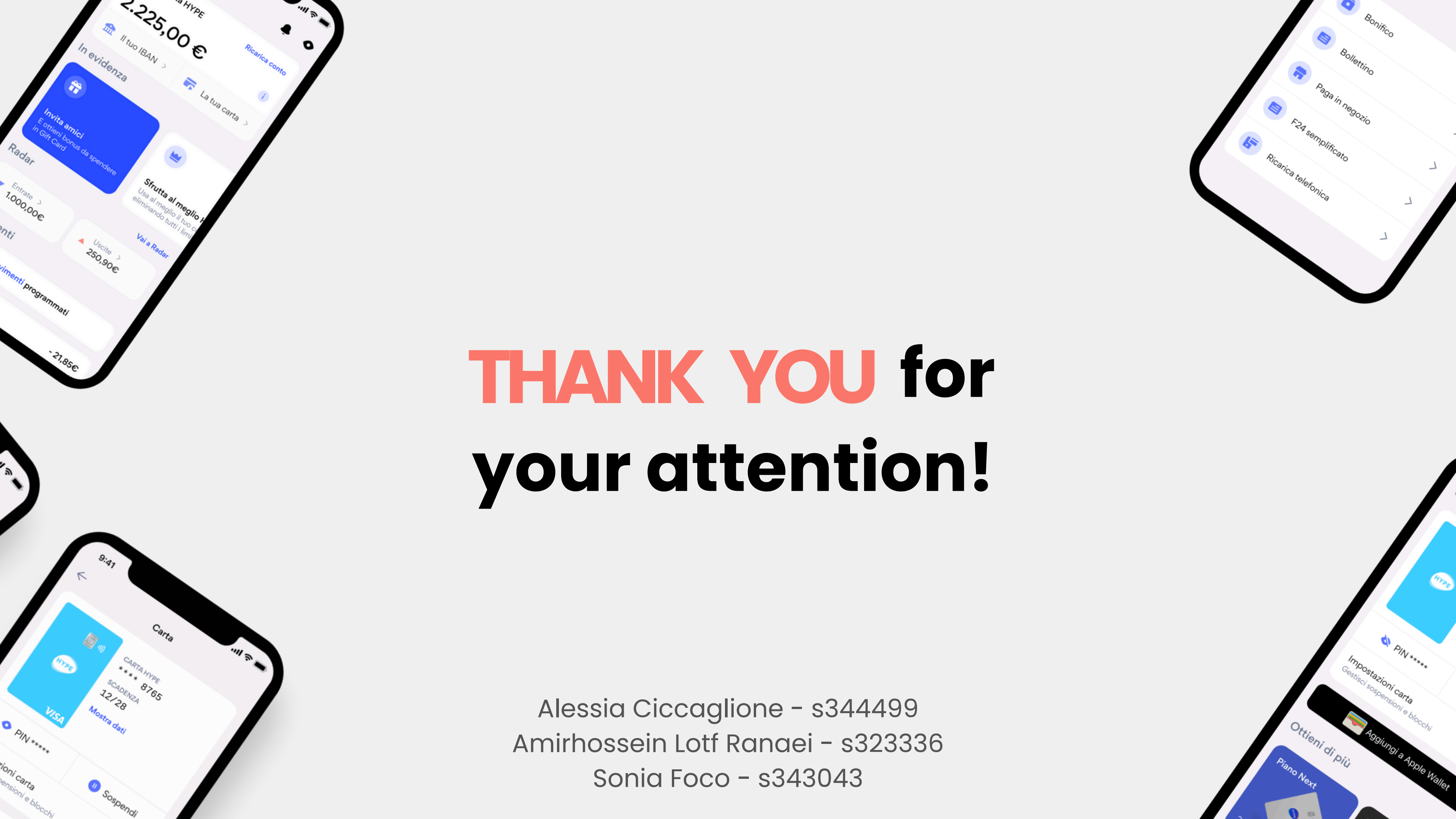


EDA, analysis and refinement of **neutral reviews**, removal of **duplicates**, execution of **stopword** analysis and conduction of general research activities.



Development of **topic-detection pipelines**, use of **BERTopic**, general research work, and organization of **code** and implementation **resources**

Stay tuned for
the final
MILESTONE...

The background features a collage of four smartphone screens displaying the HYPE app interface. The top-left screen shows the account balance of 2.225,00 € and various service buttons like 'Invita amici', 'Radar', and 'Sfrutta al meglio'. The top-right screen displays a list of services including 'Bonifico', 'Bollettino', 'Paga in negozio', 'F24 semplificato', and 'Ricarica telefonica'. The bottom-left screen shows a digital representation of a Visa HYPE card with its details and a PIN. The bottom-right screen shows the 'Impostazioni carta' (Card Settings) section with options for PIN, card suspension, and adding to Apple Wallet.

THANK YOU for your attention!

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Amirhossein Lotf Ranaei - s323336
Sonia Foco - s343043